

Final Write-Up: Time Series Forecasting of Ice Cream Production

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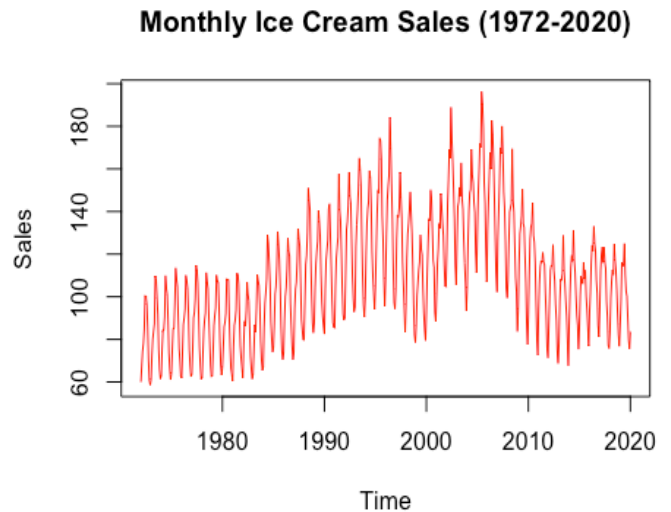
Research Question

There is a certain correlation between outside temperature and ice cream sales, but how does time of the year effect the production and sales of ice cream? So, how does the month of the year affect ice cream sales in the United States, and how can we accurately forecast future monthly production using time series modeling techniques? Can this help with maximizing profit and minimizing oversupply?

Overview and Analysis of Dataset

The dataset contains monthly U.S. ice cream production data from January 1972 to January 2020. The variable of interest monthly ice cream sales. The data clearly exhibit both a strong seasonal pattern and an upward long-term trend. As expected, sales peak during summer months and dipping during winter months.

- *June sales are 14% above average*
- *December sales are 31% below average*



```
#Viewing ratios:
sales.june<-window(sales.ts, start=c(2019,6), freq=TRUE) #seems to have highest sales
sales.dec<-window(sales.ts, start=c(2019,12), freq=TRUE) #seems to have the lowest sales
June.ratio<-mean(sales.june)/mean(sales.ts)
December.ratio<-mean(sales.dec)/mean(sales.ts)

June.ratio #14% higher than average
## [1] 1.139092

December.ratio #31% lower than average
```

- ## [1] 0.6901379

The presence of periodic fluctuations indicates seasonality, while the steady increase in the mean level points to a trend component.

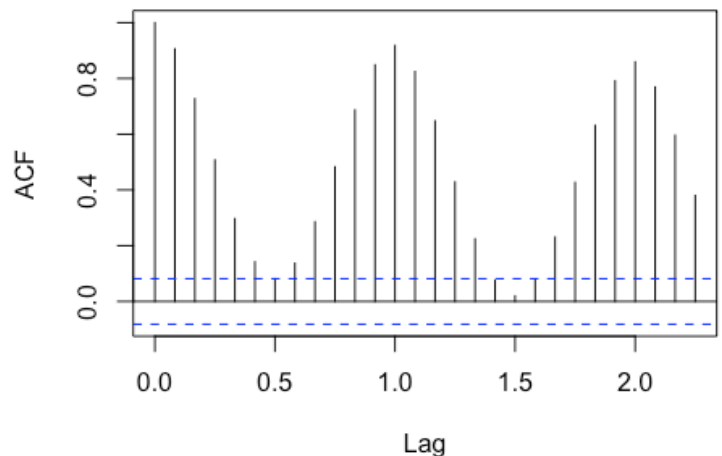
- Autocorrelation and partial autocorrelation of ice cream sales:

Very visible pattern associated with the autocorrelation of lags. This shows a strong seasonal behavior and the need for an advanced model.

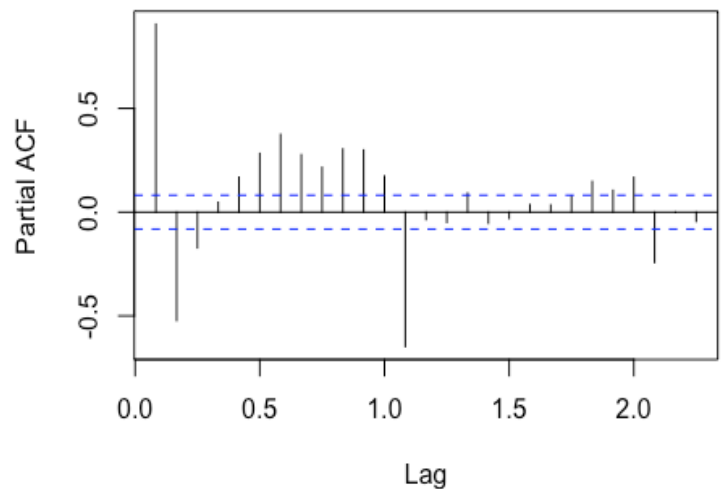
Although, it is difficult to tell what AR model would be appropriate for this dataset.

Even after decomposition the strong pattern in the autocorrelation of lags remained.

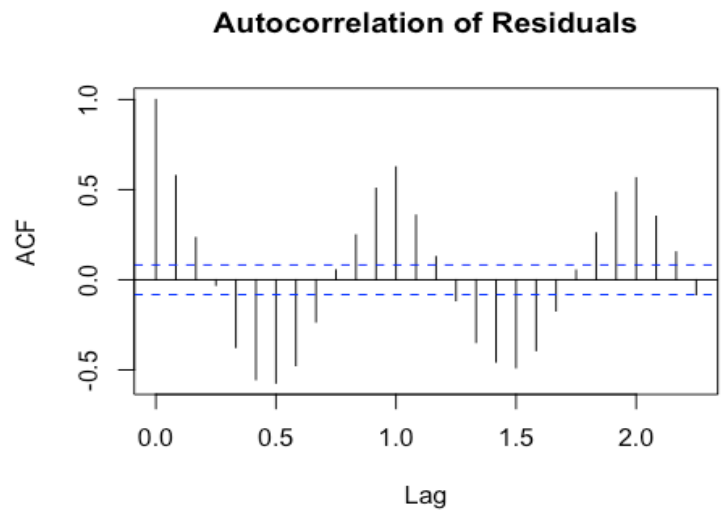
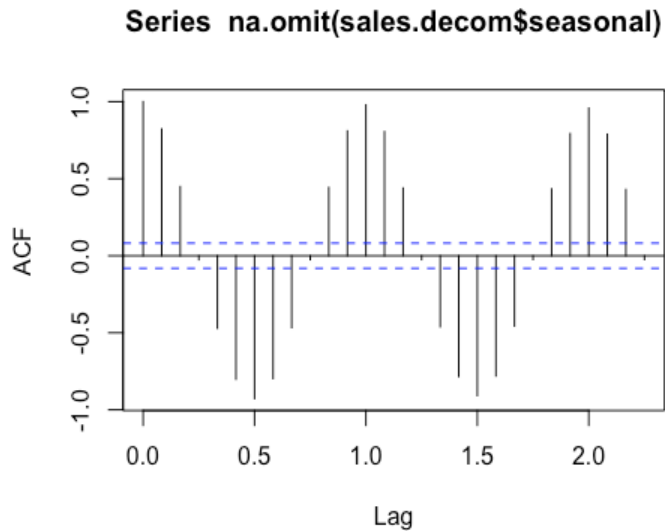
Autocorrelation for Ice Cream Sales



Partial Autocorrelation for Ice Cream Sales



Even the residuals displayed a high autocorrelation. (as seen below)



Models That Didn't Work

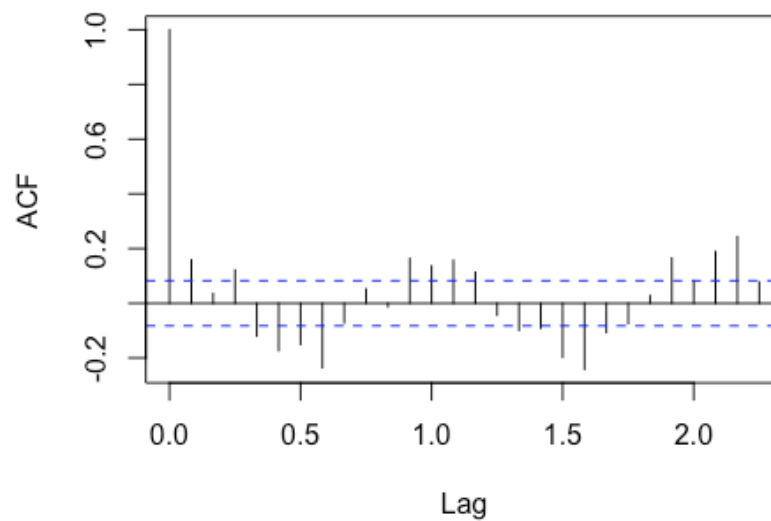
- Exponential Smoothing and Holt-Winters:
 - These models capture trend and seasonality but performed poorly because the dataset needed a more complex model. These models produced very high least-squared-errors: 21,519 and 10,755 and still the residuals of these models were still autocorrelation.

```
sales.ep$SSE  
## [1] 21519.24
```

```
sales.hw$SSE
```

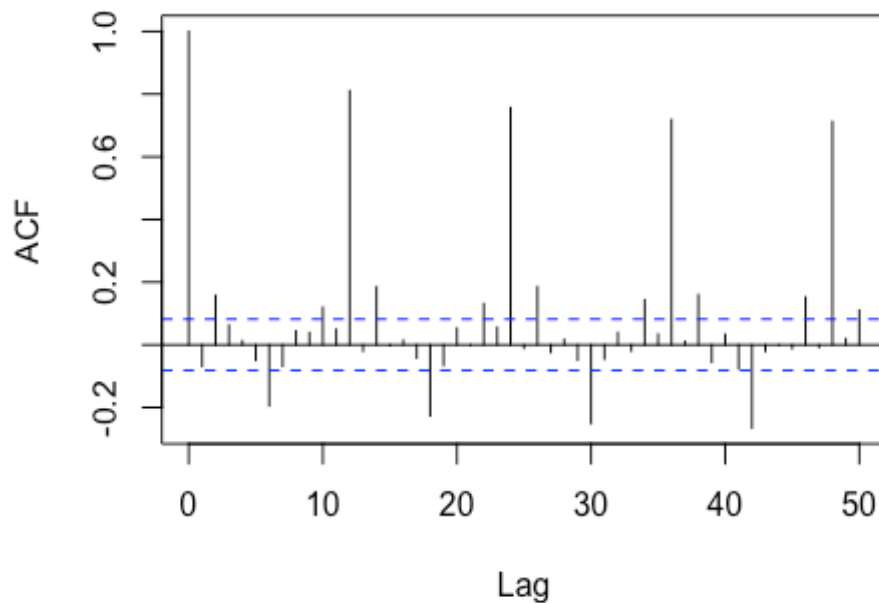
```
## [1] 10755.27
```

Holt-Winters Residuals



- Autoregressive model:
 - R produced AR(25) as the best fit autoregressive order. This is messy and shows that an autoregressive model is not the best model for this dataset. You can see in the previous ACF and PACF that the autocorrelation of values forms a complex pattern that AR can't handle. This model also had highly autocorrelated residuals.

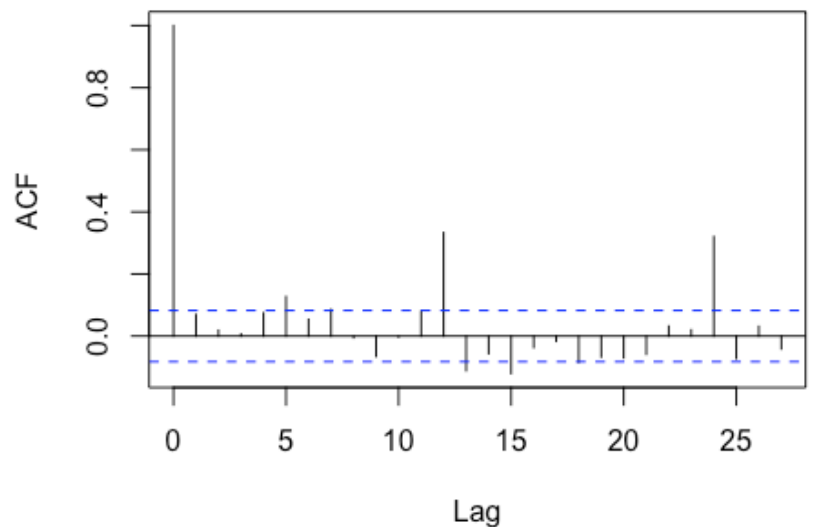
Correlation of Residuals



- Linear regression and Harmonic terms:
 - Multiple linear models with harmonic terms were tested and each one had a very large AIC (4000+). Even models with quadratic and cubic terms were used, which did have a lower AIC but not a significant amount. A log transformation was also tested; this produced an AIC of -1143.39, but the residuals were still autocorrelated.

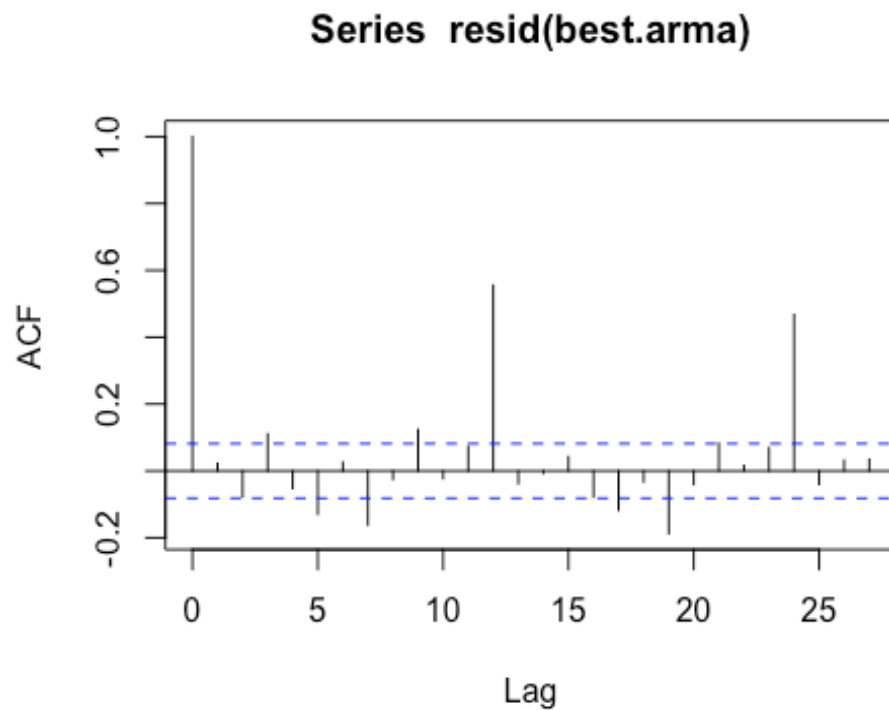
- ARIMA:
 - The best ARIMA model order was (2,0,2) and the lowest AIC was -1958.33. This was the best AIC so far, but the residuals

Series res.log.ar\$res[-(1:12)]



were still visibly autocorrelated.

```
best.aic
## [1] -1958.333
best.order
## [1] 2 0 2
```



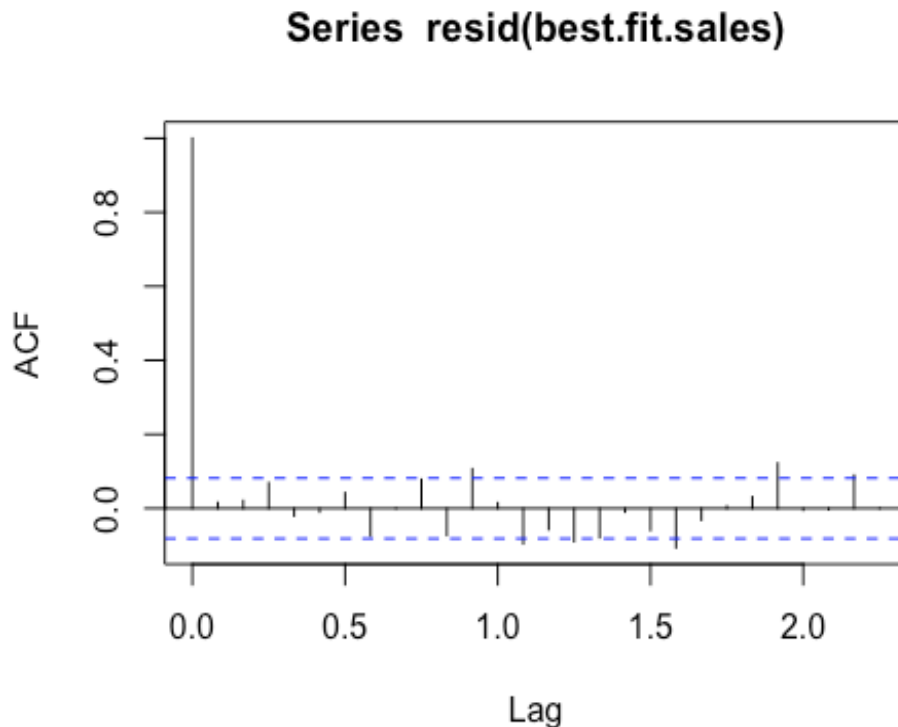
Best Model and Prediction

The best model produced was with SARIMA on Log-Transformed Data.

Since the log transformation seemed to be successful with the linear regression models, it was used here as well.

The best SARIMA model found had the order $(2,1,0)(1,0,1)[12]$ which produced an AIC of -2238.88. The residuals for this model also appear to be white noise.

```
best.arima.sales [[3]] #(210)(101)
## [1] 2 1 0 1 0 1
best.arima.sales[[1]] #-2238; really good aic
## [1] -2238.878
```



Now that the best model has been found to model ice cream sales against time, we can make a prediction.

In order to accurately predict the sales, we need to use the empirical correction factor to reverse the log transformation, so the values are appropriate for a prediction.

```
pred <- predict(best.fit.sales, 12)

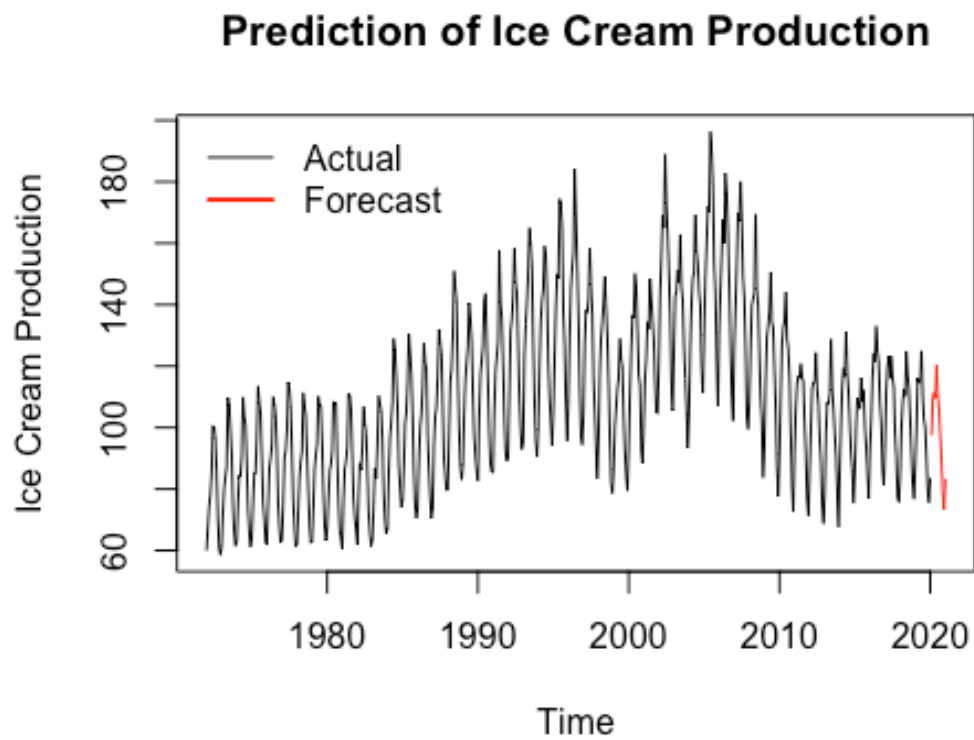
#empirical correction factor
corrected.forecast <- exp(pred$pred + 0.5 * pred$se^2)

forecast.ts <- ts(corrected.forecast, start = c(2020, 2), frequency = 12)
```

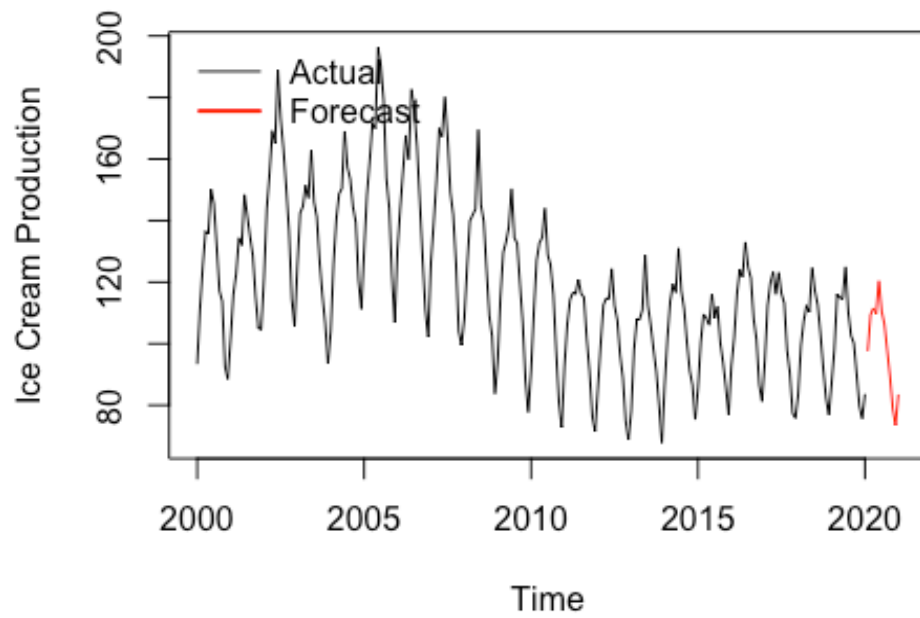
Using forecast.ts in code:

```
ts.plot(
  window(sales.ts, start = 1972),
  forecast.ts,
  lty = 1:1,
  col = c("black", "red"),
  lwd = c(1, 2),
  main = "Prediction of Ice Cream Production",
  ylab = "Ice Cream Production"
)
```

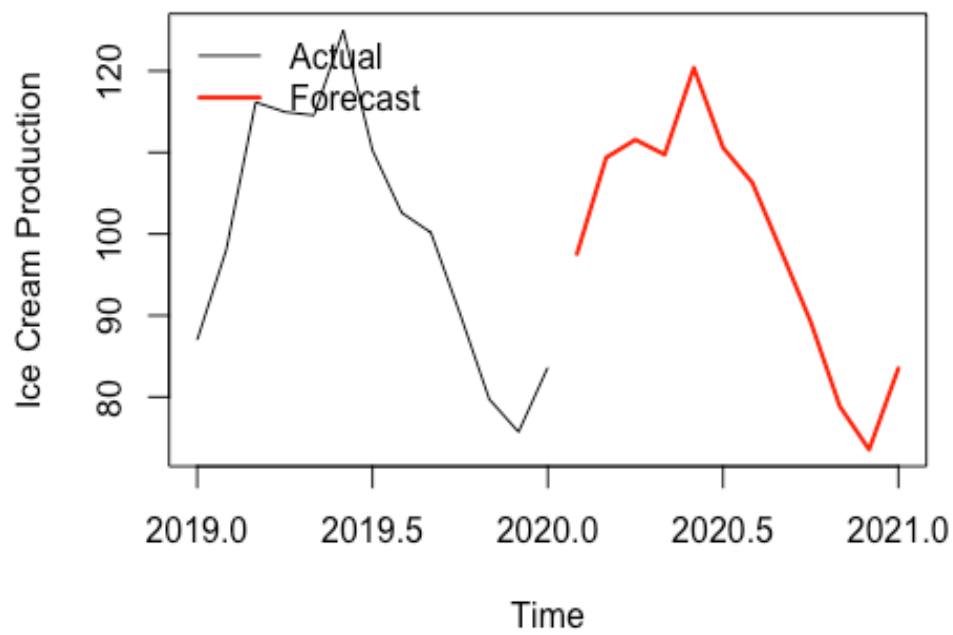
Below is the prediction plotted; each graph zooms in more on the prediction for easier analysis.



Prediction of Ice Cream Production



Prediction of Ice Cream Production



Conclusion

The SARIMA(2,1,0)(1,0,1)[12] model on log-transformed data provided the most reliable forecast for monthly ice cream production. Seasonal cycles are the dominant feature, with summer months consistently showing the highest production. Earlier models like Holt-Winters and basic AR models failed to remove residual correlation, making them less appropriate.

By thoroughly comparing models using AIC, residual analysis, and transformation techniques, the final approach successfully balanced accuracy and interpretability. This kind of modeling can be valuable for inventory planning and market analysis in seasonal industries like food production.

When looking at the most zoomed in graph of the prediction, you can see that the model predicts that the sales of ice cream peak right around the middle of the year, which would be June. It dips in the winter months; from the plot it appears the lowest month for ice cream sales appears to be late November/early December.

This model shows specific months where ice cream sales are lower or higher rather than just seasons, which can be important for a company's concern: maximizing profit and avoiding oversupply/losing money.